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Glossary

Ampulla – The widened opening for each of the semicircular canals, containing sensory innervations.

Depolarization – The loss of electric potential difference between the inside and outside the nerve cell due to a change in permeability and migration of sodium ions to the interior.

Neurotransmitters – Also known as chemical messengers, are chemicals that transmit signals across a neuromuscular junction, from one neuron to another "target" neuron, muscle cell, or gland cell.

Nissl's granules – Endoplasmic reticulum with free ribosomes, found in the cytoplasm of neuronal cell body, but absent in the axon. These are the site of protein synthesis.

Nodes of Ranvier – Periodic gap in the insulating sheath (myelin) on the axon of certain neurons that serves to facilitate the rapid conduction of nerve impulses.

Properioception – The ability to sense stimuli arising within the body regarding position, motion and equilibrium.

Schwann cells – It is also called neurilemma cell, produce the myelin sheath around neuronal axons in the peripheralneural system. Schwann cells are named after German physiologist Theodor Schwann, who discovered them in the 19th century.

Septum pellucidum – Located in the midline of the brain, between the two cerebral hemispheres. It separates the lateral ventricles I and II.

Threshold stimulus – The minimum stimulus needed to achieve an action potential.

Evaluation

Choose the followings:

1 Which structure in the ear converts pressure waves to action potentials?

a. Tympanic membrane

b. Organ of Corti

c. Oval window

d. Semicircular canal

2. Which of the following pairings is correct?

a. Sensory nerve – afferent

b. Motor nerve - afferent

c. Sensory nerve – ventral

d. Motor nerve – dorsal

3. During synaptic transmission of nerve impulse, neurotransmitter (P) is released from synaptic vesicles by the action of ions (Q). Choose the correct P and Q.

a. P = Acetylcholine, Q = Ca++

b. P = Acetylcholine, Q = Na+

c. P = GABA, Q=Na+

d. P = Cholinesterase, Q = Ca++

4. Examine the diagram of the two cell types A and B given below and select the correct option.

a. Cell-A is the rod cell found evenly all over retina

b. Cell-A is the cone cell more concentrated in the fovea centralis

c. Cell-B is concerned with colour vision in bright light